REFERENCE NOTE / MODEL IMPLEMENTATION

Rain-clutter returns for 77 GHz FMCW radar

Levels, Doppler spectra, fluctuations, and source traceability

Prepared 14 October 2026 | Version 1.0

Purpose. Add rain *returns* to an existing radar range-performance model. Propagation attenuation is assumed to be implemented already; this note supplies a compatible clutter-power and spectral model rather than another attenuation law.

> **Recommended baseline.** Compute volume backscatter from a drop-size distribution and water-drop Mie scattering at the actual radar frequency. Integrate it through the radar's two-way angular and range responses. Construct the Doppler spectrum from drop motion, wind, platform motion, viewing geometry, and an explicit unresolved-motion width. Then apply the actual Doppler filters. [R1][R4][R5][R8][R10]

The available literature supports these ingredients, but does not establish one universal "77 GHz rain coefficient plus fixed Doppler bandwidth." Direct automotive studies are particularly useful for checking *detections*; 94 GHz weather-radar work provides more detailed spectral methods. The two kinds of evidence must not be conflated. [R1][R2][R8]

Provenance. This document organizes the preceding research discussion for personal reference and a coding-agent handoff. The numerical examples were independently recomputed during export. Published results, engineering adaptations, and calculated examples are identified separately. An additional 80 GHz roadside-radar reference, found during link checking, is included as supplementary reading and is not used to set the baseline numbers. [R13]

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1 Evidence and how to use the sources

The central distinction is between **physical backscatter**, **complex radar signals**, and **detections after sensor processing**. An empirical histogram from a detection list is not automatically the amplitude law of the receiver-input clutter.

Source group	Contribution and qualification
Gourova et al. (2017) [R1]	Direct 77 GHz rain-tunnel results with a Continental ARS300; range, RCS, and velocity detections, alongside a Mie/Marshall–Palmer volume model. The observations are processed detections, not a generic I/Q model.
Gupta et al. (2026) [R2]	Continental ARS430 measurements at 16, 32, 66, and 98 mm/h; inverse-gamma and inverse-Weibull reflectivity fits. Detection thresholds, processing, and spatial selection affect the distributions.
Bertoldo et al. (2018) [R3]	A compact, Mie-derived 77 GHz equivalent-reflectivity/rain-rate fit. Useful as a cross-check, not as a universally validated measured clutter law.
Mätzler; Liebe; Marshall–Palmer [R4][R5][R6][R7]	Traceable physical ingredients: water permittivity, Mie backscattering efficiency, and an exponential drop-size distribution. The Mätzler reports include MATLAB routines.
Tsikoudi et al. (2025); Tridon & Battaglia (2015) [R8][R10]	Spectral construction from scattering and fall speed, air-motion broadening, and stochastic signal generation. Reuse the method, but recompute scattering and geometry for 77 GHz.
Myagkov et al. (2025) [R9][R11]	Measured 94 GHz spectral/polarimetric behavior and drop-shape limitations; a linked low-turbulence dataset. Not a direct automotive calibration dataset.
Su et al. (2025) [R13]	Supplementary measured-clutter study at 26.9 and 80 GHz, with range-dependent compound-Gaussian modeling and performance assessment. Useful for extending the fluctuation model beyond a homogeneous Gaussian baseline.

Reading order for an implementation

Read [R1] for the automotive volume-clutter formulation; [R4]–[R7] for mean backscatter; and [R8], [R10] for spectral construction. Use [R2] and [R13] to plan validation of statistical assumptions and processed outputs. Consult [R9] before extending the model to polarization-sensitive scattering.

The reference register in Section 7 contains clickable, written-out URLs, source locators, and access notes. A source marked “PDF retrieved” was accessible through the browsing tool during export; this is not a guarantee that every future automated downloader will be accepted by the host.

2 Mean rain-clutter level

2.1 Start with volume backscatter, not a fixed rain RCS

Use the monostatic volume backscattering coefficient

$$\boxed{\eta(f_0, T, \mathcal{R}) = \int_0^\infty N(D;\mathcal{R})\,\sigma_b(D; f_0, T)\,\mathrm{d}D.} \tag{1}$$

Here $N(D)$ is the number of drops per unit volume per unit diameter, and σ_b is the *monostatic radar cross section* of a single drop. Consequently η has units $\mathrm{m}^2/\mathrm{m}^3 = \mathrm{m}^{-1}$. Equation (1) is the physical backscatter integral used by the cited models. [R1][R4]

A traceable initial drop-size distribution (DSD) is Marshall–Palmer:

$$N(D;\mathcal{R}) = 8000 \exp\left[-4.1\mathcal{R}^{-0.21} D\right], \tag{2}$$

with D in **mm**, rainfall rate $\mathcal{R}$ in **mm/h**, and N in $\mathrm{m}^{-3}\,\mathrm{mm}^{-1}$. It is an empirical baseline, not a claim that a rain rate uniquely fixes every real drop population. [R7]

At 77 GHz, use frequency-specific **Mie scattering**, rather than the Rayleigh D^6 approximation across the entire drop population. If a Mie routine returns backscattering efficiency Q_b, the required RCS is

$$\sigma_b(D) = \frac{\pi D_\mathrm{m}^2}{4} Q_b(D), \tag{3}$$

where D_m is the diameter in metres. Use Q_b, *not* total scattering or extinction efficiency. Mätzler's Mie report provides the coefficient definitions and test examples. [R1][R5]

For water permittivity, Liebe, Hufford, and Manabe (1991) is a primary temperature- and frequency-dependent reference. Mätzler's rain report implements that variant in Appendix A1, `epswater`. The 1991 and 1993 versions differ, so record which is being used. [R4][R6]

2.2 Illustrative mean levels, numerically checked

The following numbers are **calculations for this note, not measured data or a table transcribed from a paper**. Assumptions are: $f_0 = 77$ GHz, $T = 10\,^\circ$C, unmodified Marshall–Palmer DSD, spherical drops, Liebe-1991 permittivity, and diameter integration from 0.001 to 10 mm. The source chain is [R4]–[R7].

Rain rate (mm/h)	**Calculated** η (m^{-1})	$10\log_{10}[\eta/(1\,\mathrm{m}^{-1})]$ (dB)
1	1.05×10^{-4}	−39.8
5	3.94×10^{-4}	−34.0
10	6.43×10^{-4}	−31.9
25	1.16×10^{-3}	−29.4
50	1.75×10^{-3}	−27.6
100	2.58×10^{-3}	−25.9

These values reproduce the earlier discussion to the quoted precision. The companion numerical-check script uses independently written Mie equations and verifies a published single-sphere test,

the small-particle Rayleigh limit, and diameter-grid convergence. It is a reproducibility aid, not a validated full radar simulator.

Treat the rain-rate parameterization consistently. The unmodified Marshall–Palmer DSD need not give exactly the specified rain rate when its water flux is recomputed with a separate terminal-speed law. Do not silently renormalize the DSD and still expect the table above to remain unchanged; Mätzler also discusses this distinction. [R4]

Normalization checks. Monostatic RCS is not the same as differential cross section per steradian; reconcile any 4π convention explicitly. Keep mm-based DSD integration separate from metre-based drop area. Mätzler's rain-coefficient routines report km^{-1}, whereas the radar equations here require m^{-1}. Preserve the time-harmonic sign convention when passing complex permittivity or refractive index to a Mie library. [R4][R5]

2.3 A compact published cross-check

Bertoldo et al., equation (5), give the Mie-derived 77 GHz fit

$$Z_{e,77} = 130\,\mathcal{R}^{0.65}, \tag{4}$$

with linear equivalent reflectivity in $\mathrm{mm}^6/\mathrm{m}^3$. Their equation (2) defines the equivalent-reflectivity normalization. A consistent conversion is

$$\eta = \frac{\pi^5 |K_{\mathrm{ref}}|^2}{\lambda^4}\, 10^{-18} Z_e, \tag{5}$$

where K_{ref} is the dielectric reference factor used to define Z_e, and λ is in metres. Check the source's water-factor convention rather than assuming a low-frequency constant. [R3]

Use this fit as a comparison, not a replacement for the integral. For example, at 10 mm/h, the familiar expression $200\mathcal{R}^{1.6}$ is 11.4 dB above $130\mathcal{R}^{0.65}$ under the same normalization. That difference is calculated directly from the two expressions; it illustrates the danger of transferring a classical-band reflectivity law unchanged. [R3]

2.4 Convert volume backscatter to a radar-cell return

The basic equivalent-RCS formulation is

$$\sigma_{c,\mathrm{eq}} = \eta V_{\mathrm{eff}}. \tag{6}$$

Gourova et al. explicitly use this volume-clutter approach. Their sensor produced many rain detections within about 2–3 m; the main RCS histogram population was roughly −30 to −15 dBsm. These are sensor- and processing-specific observations, not universal range limits or an input-clutter distribution. [R1]

For implementation, the following **engineering extension** replaces a hard-edged resolution cell with the actual beam and range-filter weighting. For homogeneous rain, far-field operation, and a sufficiently narrow range response,

$$V_{\text{eff}}(r) \simeq r^2 \Omega_2 \Delta r_{\text{eff}}, \tag{7}$$

$$\Omega_2 = \int g_t(\Omega) g_r(\Omega)\, \mathrm{d}\Omega, \qquad \Delta r_{\text{eff}} = \int |h_r(\delta r)|^2\, \mathrm{d}\delta r. \tag{8}$$

The patterns g_t, g_r are normalized *power* patterns. The range-filter amplitude response h_r is normalized to unity at its peak. These integrals, rather than displayed bin spacing, define the effective clutter volume.

For identical Gaussian transmit/receive beams with one-way *full* half-power widths θ_a, θ_e in radians, small-angle integration gives

$$\Omega_2 = \frac{\pi \theta_a \theta_e}{8 \ln 2}. \tag{9}$$

For an unwindowed FMCW range transform with sampled sweep bandwidth B, $\Delta r_{\text{eff}} = c/(2B)$. Windowing changes the integral; zero-padding alone does not reduce it. For a conventionally normalized sampled window w_n, the corresponding range-width multiplier is its equivalent noise bandwidth in unpadded bins,

$$\text{ENBW}_{\text{bins}} = \frac{N \sum_n |w_n|^2}{|\sum_n w_n|^2}. \tag{10}$$

Equations (7)–(10) are the beam/filter-weighted formulation proposed here, derived by integrating the normalized responses.

The mean clutter power, *before Doppler filtering*, becomes

$$\boxed{P_c(r) \simeq \frac{P_t G_{t0} G_{r0} \lambda^2}{(4\pi)^3 r^4 L_{\text{sys}}} \eta V_{\text{eff}}(r)\, A_2(r).} \tag{11}$$

Here G_{t0}, G_{r0} are peak power gains, $L_{\text{sys}} \geq 1$ is the remaining system loss, and $A_2(r) \in [0, 1]$ is the existing **two-way power-transmission factor**. This is the radar equation in [R1], with explicit loss, attenuation, and response weighting; do not apply the rain attenuation twice.

Calculated example. For identical $10° \times 10°$ Gaussian beams, $\Delta r_{\text{eff}} = 0.15$ m, and $r = 50$ m, the model gives $V_{\text{eff}} = 6.47\,\text{m}^3$. At 25 mm/h, the value of η above gives

$$\sigma_{c,\text{eq}} = 7.49 \times 10^{-3}\,\text{m}^2 = -21.3\ \text{dBsm}.$$

This is integrated over the whole rain Doppler spectrum, *not* the clutter RCS in every Doppler bin.

Under these homogeneous, fixed-beamwidth and fixed-resolution assumptions, $P_c \propto r^{-2}$ before attenuation, while a fixed-RCS point-target return scales as r^{-4}. Thus clutter may become more important relative to the target even while its absolute power falls. This follows from equations (7) and (11) and agrees with the volume-clutter discussion in [R1].

3 Doppler spectrum and coherence

3.1 Project drop motion onto the line of sight

The spectral architecture is supported by the weather-radar literature: weight the drop population by backscatter, map motion into Doppler, and account for unresolved air-motion broadening. [R8][R10] The kinematics below are the proposed adaptation to an automotive radar.

Let $\hat{\mathbf{u}}$ point from the radar toward a drop and let $\hat{\mathbf{z}}$ point upward. Define positive radial velocity as receding. With mean air velocity $\mathbf{U}$, platform velocity $\mathbf{V}_{\text{radar}}$, and downward terminal speed $v_T(D)$,

$$\boxed{v_0(D, \hat{\mathbf{u}}) = (\mathbf{U} - \mathbf{V}_{\text{radar}})\cdot\hat{\mathbf{u}} - v_T(D)\,\hat{\mathbf{z}}\cdot\hat{\mathbf{u}}.} \tag{12}$$

All vectors must be expressed in the same coordinate frame. The model Doppler sign will be defined to match this receding-positive convention; the radar's I/Q or beat-frequency convention may require a sign change at the interface.

Tsikoudi et al., equation (7), give the following piecewise fall-speed parameterization. Here D_{m} is in *metres* and speed is in m/s:

$$v_T(D_{\text{m}}) = \begin{cases} 1.2\times10^{8}(D_{\text{m}}/2)^2, & D_{\text{m}} < 0.11\times10^{-3}, \\ 8333(D_{\text{m}}/2) - 0.0833, & 0.11\times10^{-3} \le D_{\text{m}} \le 0.86\times10^{-3}, \\ 9.65 - 10.3e^{-600D_{\text{m}}}, & D_{\text{m}} > 0.86\times10^{-3}. \end{cases} \tag{13}$$

The optional air-density multiplier is $(\rho_0/\rho)^{0.4}$. In the rain-sized branch, the exponential is equivalently $e^{-0.6D}$ for D in mm. Do not extrapolate that branch to zero diameter. [R8]

Horizontal-looking radar. At exactly horizontal incidence, vertical fall speed has no radial component. A forward-looking automotive radar must not be assigned the several-m/s downward Doppler centroid of a vertically pointing weather radar. Finite elevation beamwidth, wind, platform motion, and unresolved dynamics can nevertheless broaden or shift the spectrum. These statements follow directly from equation (12).

The platform term should remain unless the signal-processing chain explicitly compensates it. A moving vehicle's rain-clutter centroid generally need not lie at zero in sensor coordinates.

3.2 A normalized spectral model

For homogeneous rain and spherical-drop scattering, use

$$\boxed{p_v(v) = \frac{1}{\eta\Omega_2}\int_{\Omega} g_t g_r \int_{D} N(D)\sigma_b(D)\,\varphi_{\sigma_t}[v - v_0(D, \hat{\mathbf{u}})]\;\mathrm{d}D\,\mathrm{d}\Omega.} \tag{14}$$

Here φ_{σ_t} is a unit-area Gaussian with line-of-sight velocity standard deviation σ_t, representing unresolved air-motion broadening. Consistent diameter and angular integration makes $\int p_v(v)\,\mathrm{d}v = 1$.

The scattering-weighted motion mapping and Gaussian broadening follow [R8], [R10]. The explicit beam integration and platform term are the proposed engineering adaptation. It avoids the singular diameter-to-Doppler inversion that arises when a vertical-weather-radar transformation is applied at horizontal incidence.

Rain rate is not enough to specify σ_t. Keep it as a scenario input or a calibrated parameter. A published "spectral width" may already include wind shear, beam effects, or processing; do not insert it as pure turbulence and then add the same mechanisms again. [R10]

At 77 GHz, using the exact speed of light, $2/\lambda = 513.6887$ Hz/(m/s). For the chosen model sign convention,

$$p_f(f_D) = \frac{\lambda}{2}\,p_v\!\left(\frac{\lambda f_D}{2}\right), \qquad S_c(f_D; r) = P_c(r)\,p_f(f_D). \tag{15}$$

Thus S_c has units W/Hz and integrates to the total clutter power. The Jacobian $\lambda/2$ is necessary to preserve that normalization.

3.3 Geometry-induced width: an illustrative calculation

For horizontal boresight, identical Gaussian elevation patterns, no wind, no platform motion, and a *symmetric, untruncated rain-filled beam*, the small-angle model gives

$$\sigma_{v,\mathrm{fall}} \simeq \frac{\theta_e}{4\sqrt{\ln 2}}\sqrt{\langle v_T^2\rangle_b}, \qquad \langle v_T^2\rangle_b = \frac{\int N(D)\sigma_b(D)v_T(D)^2\,\mathrm{d}D}{\eta}. \tag{16}$$

This is a derivation for the stated geometry, not a measured rain-clutter bandwidth.

At 25 mm/h, 77 GHz, and 10 °C, the same numerical model with equation (13) gives a backscatter-weighted RMS fall speed of 5.113 m/s. For a 10° elevation beam,

$$\sigma_{v,\mathrm{fall}} = 0.268\ \mathrm{m/s}, \qquad \sigma_f = 137.7\ \mathrm{Hz}.$$

The idealized spectrum is centred at zero. These are geometry-only *RMS* widths, not full widths or universal automotive values. In an actual low-mounted radar, the ground may truncate the lower part of the beam; then the symmetric example and its zero centroid need not apply.

For a reduced model, a computed spectrum can be moment-matched to a Gaussian. With $R_c(\tau) = \mathbb{E}[c(t+\tau)c^*(t)]$, its autocorrelation is

$$R_c(\tau) = P_c \exp(j2\pi\bar{f}_D\tau)\exp(-2\pi^2\sigma_f^2\tau^2). \tag{17}$$

This derived relationship keeps spectral width and coherence consistent instead of choosing an independent decorrelation time.

4 Fluctuation statistics and empirical validation

For a homogeneous volume with many independent contributors, a useful baseline is zero-mean circular complex Gaussian clutter with the specified spectrum. Its envelope is Rayleigh and its instantaneous power is exponential. **Gaussian I/Q statistics do not require a Gaussian Doppler spectrum.** The published weather-signal methods generate stochastic complex spectra and corresponding I/Q realizations. [R8][R10]

For simulation, generate complex Gaussian spectral coefficients with variances set by S_c and inverse-transform with a checked normalization. Generating independent samples at every chirp discards the desired temporal correlation. The transform scaling must make the time-domain variance equal to P_c.

Gupta et al.'s inverse-gamma and inverse-Weibull fits apply to reflectivity inferred from *detection lists*. The paper acknowledges detection thresholds, range-dependent sensitivity, and proprietary processing, and does not establish cross-sensor universality. Use these fits for comparable post-detection outputs, rather than automatically as a raw receiver-input amplitude law. [R2]

Small volumes and texture

For small effective volumes, test the many-contributor assumption. A discrete-drop alternative samples the DSD, positions, and velocities, then sums complex returns. The spatial point process and clustering assumptions must be documented separately; a DSD alone does not define them. This is a recommended modeling extension, not a parameter set measured by the cited automotive studies.

As a supplementary validation lead, Su et al. study measured roadside clutter at 26.9 and 80 GHz and fit range-varying compound-Gaussian models. Their preferred texture distributions depend on

frequency and rainfall category. This is evidence worth examining before assuming homogeneous Gaussian clutter universally, but it does not justify transferring their fitted parameters unchanged to a different sensor. It was found during export and is not used in this note's numerical baseline. [R13]

Polarization and a historical spectral trap

For polarization-sensitive work, spherical Mie scattering is only a baseline. The recent W-band studies show limitations in simple drop-shape/orientation approximations, especially in spectral polarimetric features. A shape- and canting-aware model warrants separate validation at the intended frequency and geometry. [R8][R9]

The older Currie, Dyer, and Hayes report covers 9.375, 35, 70, and 95 GHz amplitude statistics and *non-coherent* spectral measurements. Its detected-signal fluctuation spectra must not be treated as coherent FMCW Doppler spectra. [R12]

5 Integration with the FMCW performance model

For performance prediction, the relevant clutter quantity is usually the clutter *at the target's Doppler*, not the total rain return. Given equation (15), the mean clutter at output k of a Doppler filter is

$$\boxed{C_k(r) = \int S_c(f_D; r)\,|H_k(f_D)|^2\,\mathrm{d}f_D,} \tag{18}$$

with H_k normalized consistently with the point-target signal and thermal-noise model. This is the filter-output variance derived from the proposed input PSD.

A first performance estimate can use target output power divided by $N_k + C_k$. Detection probability and CFAR behavior require the chosen fluctuation statistics, correlations, and detector details; a mean-power ratio alone does not specify them.

1. **Total power versus Doppler-bin power.** Do not add all of P_c to every bin, or spread it uniformly unless a flat spectrum is actually intended.
2. **Resolution versus displayed spacing.** Use the range and Doppler filter responses. Zero-padding does not create smaller independent clutter volumes or new independent Doppler samples.
3. **Atmosphere versus instrument.** Apply slow-time sampling, aliasing, windows, clutter rejection, and any motion compensation explicitly. Do not duplicate these effects inside the atmospheric broadening parameter.

Scope of the separable baseline

The simple expressions assume locally homogeneous rain and a separable range/angular response, with scatterer statistics approximately stationary over the coherent processing interval. Use the actual post-beamforming receive pattern if angle processing is part of the model; do not count antenna gain twice.

Restrict the integration to rain-filled air when the beam intersects the ground, a tunnel, or another boundary. If attenuation, rain density, beam weighting, or range spreading varies materially across a cell, integrate the radar equation over the volume rather than factoring out a single r, η, and A_2. Significant range migration or coupled angle/range/Doppler processing calls for a signal-level extension. These are stated limits of the proposed implementation, not new measured claims.

6 Coding-agent implementation brief

6.1 Interfaces and staged implementation

Use SI units internally except where a source parameterization explicitly specifies mm or GHz. Keep both the model convention and the conversion at the interface visible in code.

Module	Inputs, outputs, and provenance
Microphysics	Inputs: rain-rate/DSD choice, temperature, f_0, diameter grid. Output: $N(D)$, $\sigma_b(D)$, η. Use equations (1)–(3); [R4]–[R7].
Cell power	Inputs: range, two-way beam weighting, range response, radar gains/losses, existing A_2. Output: V_{eff} and P_c. Equations (7)–(11); automotive basis [R1], weighted extension in this note.
Spectrum	Inputs: velocity vectors, beam geometry, fall-speed law, σ_t. Output: unit-area p_v, p_f, and power PSD S_c. Equations (12)–(15); [R8], [R10], and stated geometry adaptation.
Processing	Inputs: sample timing, windows, H_k, compensation and rejection filters. Output: C_k and, optionally, correlated I/Q. Equation (18); retain the existing target/noise normalization.
Validation	Compare like-for-like outputs. Use [R1], [R2] for automotive detections; inspect [R13] for richer amplitude statistics and [R9], [R11] for 94 GHz spectral-method checks.

Cache scattering as a function of diameter, temperature, and frequency when useful. Introduce alternate DSDs, texture, polarization, and signal-level simulation as separate optional extensions rather than hiding their assumptions in a fitted constant.

6.2 Minimum acceptance checks

Single-drop scattering. Reproduce the Mätzler test $m = 5+0.4j$, $x = 1$, which gives $Q_b \simeq 1.1138$. In the small-particle limit, recover $Q_b \simeq 4x^4|(m^2-1)/(m^2+2)|^2$. Reconcile the solver's imaginary-index sign convention. [R5]

Numerical convergence. Reproduce the mean-level table with its exact assumptions, then refine the diameter grid and upper bound. Verify that the retained Doppler grid captures essentially all spectral power. Do not infer physical model accuracy merely from numerical convergence.

Normalization and limiting geometry. Check $\int p_v \, \mathrm{d}v = 1$, $\int p_f \, \mathrm{d}f_D = 1$, $\int S_c \, \mathrm{d}f_D = P_c$, and generated I/Q variance P_c. With zero angular width, horizontal incidence, and no wind/motion/-turbulence, the fall-speed contribution should collapse to zero radial width. With $\sigma_t \to 0$, use a narrow-kernel or histogram limit rather than dividing by zero.

Processing checks. Verify equation (18) against Monte Carlo output variances, including aliasing and the actual windows. Validate total versus per-bin power and preserve units through dB conversions. Use at least one deterministic point-target test to check target/clutter normalization consistency.

6.3 Reproducibility and document retrieval

The accompanying handoff bundle contains this PDF and its editable LaTeX source. Machine-readable links are in `source_links.json`. The script `verify_numbers.py` produces `numerical_checks.json`. It checks the illustrative mean backscatter and geometric width only; it does not implement the complete clutter spectrum or receiver chain.

For downloading references, use each verified `pdf_url` first; otherwise use the full-text, DOI, or record URL. Follow normal redirects and confirm that a supposed PDF has a PDF signature and the expected document title. Some endpoints return HTML, authentication, or anti-bot pages instead. Record the URL, access date, version, and a file hash for any downloaded reference. Respect the source's access and reuse conditions; the bundle does not contain copies of the papers.

The data DOI in [R11] is confirmed by the publisher article, but dataset retrieval was not verified in this environment. The Cambridge reference [R2] has verified full-text HTML rather than a verified direct-PDF endpoint. These are access limitations, not missing physical ingredients for the baseline model.

> **Implementation decision.** Start with DSD + frequency-specific Mie backscatter, the weighted clutter volume, and the geometry-aware spectrum. Keep unresolved motion and non-Gaussian extensions configurable. Validate the resulting processed quantities against comparable measurements; do not claim a universal automotive calibration from this reference note alone.

7 Reference register and retrieval links

Access checked: 14 October 2026. "PDF retrieved" means the browsing tool returned a readable PDF. Standalone programmatic network requests were unavailable in the code runtime, so bulk-download compatibility was not tested. URLs are both clickable and printed as text. Stable DOI links are retained where available; access conditions can change.

[R1] R. Gourova, O. Krasnov, and A. Yarovoy (2017).

Analysis of Rain Clutter Detections in Commercial 77 GHz Automotive Radar. Proceedings of the 14th European Radar Conference, pp. 25-28.

Use: Direct automotive detections and the volume-clutter radar equation; not universal raw-I/Q statistics.

Read: Sections II-C and III-B; equations (3)-(7); Figures 3 and 5.

Access: PDF retrieved by the browsing tool (5 pages). Author-hosted ResearchGate copy; automated clients may encounter access controls. TU Delft portal confirms the publication.

DOI: https://doi.org/10.23919/EURAD.2017.8249138

PDF: https://www.researchgate.net/profile/Oleg-Krasnov/publication/323135471_Analysis_of_rain_clutter_detections_in_commercial_77_GHz_automotive_radar/links/5e568c1b4585152ce8f15da1/Analysis-of-rain-clutter-detections-in-commercial-77-GHz-automotive-radar.pdf

Record: https://research.tudelft.nl/en/publications/analysis-of-rain-clutter-detections-in-commercial-77-ghz-automoti/

[R2] S. Gupta, I. Varga, M. A. Hein, and T. Dallmann (2026).

Analysis and empirical modeling of rain clutter for automotive radar validation. International Journal of Microwave and Wireless Technologies 18(3), pp. 401-408; online 9 March 2026.

Use: Empirical fits to detected automotive rain returns; check processing, censoring, and spatial selection before reuse.

Read: Measurement setup, Methodology, and Results and discussion; inverse-Weibull and inverse-gamma models and limitations.

Access: Publisher full-text HTML retrieved. A working direct PDF download was not established during export; use the HTML or publisher download control.

DOI: https://doi.org/10.1017/S1759078726103055

Full text: https://www.cambridge.org/core/journals/international-journal-of-microwave-and-wireless-technologies/article/analysis-and-empirical-modeling-of-rain-clutter-for-automotive-radar-validation/FEC14F285F2D50E16B0B9E239E366CD4

[R3] S. Bertoldo, C. Lucianaz, and M. Allegretti (2018).

On the Use of a 77 GHz Automotive Radar as a Microwave Rain Gauge. Engineering, Technology & Applied Science Research 8(1), pp. 2356-2360.

Use: Compact 77 GHz equivalent-reflectivity/rain-rate cross-check, rather than a universal measured clutter law.

Read: Equation (2), definition of equivalent reflectivity; equation (5), Z(77 GHz) = 130 R^0.65, printed p. 2358.

Access: Publisher PDF retrieved (5 pages).

DOI: https://doi.org/10.48084/etasr.1755

PDF: https://etasr.com/index.php/ETASR/article/download/1755/pdf/4515

[R4] C. Mätzler (2002).

Effects of Rain on Propagation, Absorption and Scattering of Microwave Radiation Based on the Dielectric Model of Liebe. University of Bern, Institute of Applied Physics, Research Report 2002-10.

Use: DSD-integrated backscatter and the implementation of the Liebe-1991 water model.

Read: Sections 3.1-3.2; Appendix A1 (epswater), A3 (Mie_rain2), and A4 (Mie_rain3). Coefficient outputs use km^-1.

Access: Full report PDF retrieved (30 pages), from a third-party mirror of the original research report.

PDF: https://pi6ats.nl/rain%20effect.pdf

[R5] C. Mätzler (2002).

MATLAB Functions for Mie Scattering and Absorption. University of Bern, Institute of Applied Physics, Research Report 2002-08, June 2002.

Use: Mie coefficients, backscattering efficiency Q_b, and numerical test cases.

Read: Sections 2-3; function Mie in Section 3.3; numerical example m=5+0.4i, x=1 in Section 4.1 (printed p. 14).

Access: Full report PDF retrieved (19 pages), hosted by the Oregon Medical Laser Center.

PDF: https://omlc.org/software/mie/maetzlermie/Maetzler2002.pdf

[R6] H. J. Liebe, G. A. Hufford, and T. Manabe (1991).

A Model for the Complex Permittivity of Water at Frequencies Below 1 THz. International Journal of Infrared and Millimeter Waves 12(7), pp. 659-675.

Use: Primary dielectric-model reference; use the 1991 variant consistently with the numerical checks.

Read: Two-relaxation model; see also the executable parameterization in R4, Appendix A1.

Access: Original paper PDF retrieved (17 pages) from NTIA.

DOI: https://doi.org/10.1007/BF01008897

PDF: https://its.ntia.gov/publications/download/LiebeHuffordManabe1991.pdf

[R7] J. S. Marshall and W. McK. Palmer (1948).

The Distribution of Raindrops with Size. Journal of Meteorology 5(4), pp. 165-166.

Use: The baseline exponential drop-size distribution; an empirical population assumption, not an exact universal rainfall law.

Read: Equations (1)-(3), printed p. 165.

Access: University-hosted scanned paper PDF retrieved (2 pages). Equations on the first page were checked visually.

PDF: https://www.atmos.albany.edu/facstaff/rfovell/ATM562/marshall-palmer-1948.pdf

[R8] I. Tsikoudi, A. Battaglia, C. Ünal, and E. Marinou (2025).

Simulations of spectral polarimetric variables measured in rain at W-band. Atmospheric Measurement Techniques 18, pp. 4857-4870.

Use: Drop fall speeds, spectral forward modeling, stochastic complex-signal generation, and polarization caveats at 94 GHz.

Read: Equation (7), piecewise fall speed, printed p. 4861; Section 2.2, spectral simulation; Section 4, limitations.

Access: Open-access publisher PDF retrieved (14 pages); HTML is also available.

DOI: https://doi.org/10.5194/amt-18-4857-2025

PDF: https://amt.copernicus.org/articles/18/4857/2025/amt-18-4857-2025.pdf

Full text: https://amt.copernicus.org/articles/18/4857/2025/

[R9] A. Myagkov, T. Nomokonova, and M. Frech (2025).

Empirical model for backscattering polarimetric variables in rain at W-band: motivation and implications. Atmospheric Measurement Techniques 18, pp. 1621-1640.

Use: Measured 94 GHz spectral/polarimetric behavior and limitations of simple drop-shape approximations.

Read: Empirical scattering model, conclusions, and Code and data availability.

Access: Open-access publisher PDF retrieved (20 pages). Article links to a model supplement and the dataset in R11.

DOI: https://doi.org/10.5194/amt-18-1621-2025

PDF: https://amt.copernicus.org/articles/18/1621/2025/amt-18-1621-2025.pdf

Full text: https://amt.copernicus.org/articles/18/1621/2025/

[R10] F. Tridon and A. Battaglia (2015).

Dual-frequency radar Doppler spectral retrieval of rain drop size distributions and entangled dynamics variables. Journal of Geophysical Research: Atmospheres 120, pp. 5585-5601.

Use: Backscatter-weighted ideal Doppler spectra, air-motion broadening, and stochastic spectral realizations.

Read: Forward-model and synthetic-spectrum methodology; distinguish air broadening from drop-size spectral width.

Access: Publisher PDF retrieved (17 pages). Automated clients may be subject to publisher access controls.

DOI: https://doi.org/10.1002/2014JD023023

PDF: https://agupubs.onlinelibrary.wiley.com/doi/pdfdirect/10.1002/2014JD023023?download=true

[R11] A. Myagkov and T. Nomokonova (2025).

W-band spectra collected under low turbulence conditions. Zenodo dataset linked by R9.

Use: Potential spectral-method validation data at 94 GHz; not a 77 GHz automotive calibration dataset.

Read: Inspect the dataset description, files, metadata, and license before use.

Access: Dataset DOI and title confirmed in the publisher article/assets page. Dataset file retrieval was not verified; the Zenodo endpoint failed in the browsing environment.

DOI: https://doi.org/10.5281/zenodo.14849259

Record: https://zenodo.org/records/14849259

Dataset provenance: https://amt.copernicus.org/articles/18/1621/2025/amt-18-1621-2025-assets.html

[R12] N. C. Currie, F. B. Dyer, and R. D. Hayes (1975).

Analysis of a Radar Rain Return at Frequencies of 9.375, 35, 70, and 95 GHz. Georgia Institute of Technology, Engineering Experiment Station; DTIC ADA007254, 1 February 1975.

Use: Historical amplitude statistics and non-coherent fluctuation spectra; do not treat these as coherent FMCW Doppler spectra.

Read: Catalog abstract explicitly describes non-coherent spectral measurements.

Access: Archive catalog record and abstract found. Direct PDF/full-text download could not be revalidated during export; use the record download controls.

Record: https://archive.org/details/DTIC_ADA007254

[R13] H.-L. Su, X.-J. Zhang, Y.-H. Li, Y.-H. Wang, P.-J. Zou, and P.-L. Shui (2025).

Rain Clutter Modelling and Performance Assessment of Road Traffic Surveillance Radars. IET Radar, Sonar & Navigation, article e70008.

Use: Measured 26.9 and 80 GHz roadside clutter, range-dependent compound-Gaussian modeling, and radar-performance assessment. Not used in the baseline numerical calculations.

Read: Sections 3-5, compound-Gaussian model and measured-data evaluation; read sensor and preprocessing details before transferring fitted parameters.

Access: Open-access publisher PDF retrieved (15 pages). This supplementary reference was found during export.

DOI: https://doi.org/10.1049/rsn2.70008

PDF: https://ietresearch.onlinelibrary.wiley.com/doi/epdf/10.1049/rsn2.70008